Streaming: Exercises

Exercise

Let $\Sigma = x_1, x_2, \ldots x_n$ be a stream of n distinct integers in $[1, n+1]$. Design a streaming algorithm that find the missing number using $O(1)$-size working memory, 1 pass, and $O(1)$ time per element.

Exercise: High Probability Guarantees (median trick)

Consider a stream Σ with elements from a universe U and let F_0 be t he number of distinct elements in Σ. Suppose that you run ℓ *independent instances* of the probabilistic counting argument, obtaining ℓ estimates $\tilde{F}_0^{(j)}$ for F_0, with $1 \le j \le \ell$. Assume ℓ odd and let $\tilde{F}_0$ be the mediam value of the $\tilde{F}_0^{(j)}$'s. Show that with a suitable choice of $\ell = \Theta(\log |U|)$ we have:

$$\Pr(\tilde{F}_0 < F_0/16) \le 1/|U| \quad \text{and} \quad \Pr(\tilde{F}_0 > 16F_0) \le 1/|U|.$$

Hint: Use the theorem and the Chernoff bound.

Exercise

For a stream Σ of n items $x_1, x_2, \ldots, x_n$, consider the following algorithm:

- Run in parallel the Boyer-Moore majority algorithm and the Sticky Sampling algorithm with $\phi = 1/2$.
- Let cand be the candidate identified by the Boyer-Moore algorithm, and S the set of items returned by the Sticky Sampling algorithm.

If cand $\in S$ **then return** cand **otherwise return** null.

What probabilistic guarantees does this algorithm provide?